MILITARY SPECIFICATION

INFRARED NIGHT DRIVING KIT; PACKAGING AND PACKING OF

1. SCOPE

1.1 This specification covers the requirements for cleaning, preservation, packaging and packing of the Infrared Night Driving Kit against external causes of deterioration during shipment and storage.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawing, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-E-760-Ethyl Alcohol (Ethanol); Denatured Alcohol; and Proprietary Solvent.
UU-P-313-Paper Lens Tissue.
CCC-C-440-Cloth, Cotton, Cheesecloth.
JJJ-C-561-Cotton; Sterile and Nonsterile.
PPP-B-601-Boxes, Wood, Cleated-Plywood.
PPP-B-621-Boxes, Wood, Nailed and Lock Corner.
PPP-B-636-Box, Fiberboard.
PPP-C-843-Cushioning Material, Cellulosic.
PPP-T-60-Tape; Pressure Sensitive Adhesive, Waterproof-For Packaging and Sealing.

MILITARY

MIL-P-116-Preservation, Methods of.
MIL-B-117-Bags, Interior Packaging.
MIL-P-130-Paper, Wrapping, Laminated and Creped.
MIL-P-1755-Powder, Talcum, Plain and Borated.
MIL-G-10924-Grease Automotive and Artillery.
MIL-P-13988-Paper, Lens, Tissue, Anti-tarnish, Wrapping.

STANDARDS

MIL-STD-105-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129-Marking for Shipment and Storage.

DRAWINGS

Bureau of Ships

E1940895-Infrared Kit.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Infrared night driving kit. - The infrared night driving kit shall consist of the parts listed on Drawing E1940895.

3.2 Materials and methods. - The materials, methods and procedures used in the packaging and packing of items shall be as specified herein. Materials and methods not covered by this specification shall be suited to the intended use and subject to the approval of the contracting officer. Unless otherwise specified, the contractor shall provide certification of adherence to the required material specifications.

3.2.1 Tape. - Pressure sensitive tape shall conform to type III of Specification PPP-T-60.

3.3 Pilot pack (level A and B packing only). - Prior to quantity production, a sample pilot pack (see 4.2) shall be submitted for approval to determine conformance with the requirements of this specification.

3.4 Preservation and packaging. - Preservation and packaging shall be level A, B or C as specified (see 6.1).

3.4.1 Level A. -

3.4.1.1 Cleaning. -

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3.4.1.1.1 Optics.

3.4.1.1.1.1 The following materials and equipment shall be used in the cleaning of optics:

Materials and Equipment	Specification
Alcohol, ethyl	Grade II of Specification O-E-760
Brush, camel's hair	
Cheesecloth	Specification CCC-C-440
Cloth, batiste	
Cotton, absorbent	Grade A, class I of Specification JJJ-C-561
Paper, lens tissue	Specification UU-P-313
Sticks, orange or equal	
Syringe, rubber	

3.4.1.1.1.2 Cleaning the exposed optics of the infrared viewer, the lens of the infrared lamps and four spare lamp lenses shall be accomplished in the following manner:

(a) Loose particles of dust shall be removed from the surface of the optic with a camel hair brush.

(b) The surface to be cleaned shall be wiped with a circular motion using a cloth saturated with alcohol. The surface shall be dried with a clean wiping cloth.

(c) A swab shall be made by wrapping one piece of lens tissue paper around the end of an orange stick or equal. One or two drops of alcohol shall be added on the tip of the swab. The swabbing operation shall be in a circular motion beginning at the center of the optic. A light downward pressure shall be exerted on the swab and the cleaning radius shall be increased until the entire surface has been swabbed. This operation shall be repeated until no trace of dirt, lint or smears are perceptible.

(d) A rubber syringe shall be used as a bellows to remove any residual dust or lint that may remain on the cleaned surface.

(e) Immediately after cleaning, the infrared viewer optics and infrared lamp lenses shall be covered with lens tissue paper conforming to Specification MIL-P-13988.

3.4.1.1.2 Other surfaces. - Cleaning process C-1 of Specification MIL-P-116 shall be used to clean all other surfaces of infrared night driving kit components including metal stowage box.

3.4.1.2 Drying. - Immediately after cleaning, the items shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

3.4.1.3 Unit preserving and packaging. - Unit preserving and packaging shall be as specified herein.

3.4.1.3.1 Infrared night driving lamps. - The lens tissue paper, conforming to Specification MIL-P-13988, shall be taped securely over the lenses with pressure sensitive tape (see 3.2.1). The connectors of the lamps' electrical cables shall be sealed with pressure sensitive tape (see 3.2.1). The threads of the lamp mounting sockets shall be coated with automotive and artillery grease conforming to Specification MIL-G-10924. The lamps shall be mounted securely in place in the stowage box. The cables of each lamp shall be immobilized by taping to the opposite lamp.

3.4.1.3.2 Viewer mounting brackets. - The unpainted surfaces of the two viewer mounting brackets shall be coated with automotive and artillery grease conforming to Specification MIL-G-10924. The brackets shall be placed in the stowage box as shown in figure 1. Immobilize the second bracket with celluloic cushioning material, conforming to Specification PPP-C-843. Position the divider plate in the stowage box.

3.4.1.3.3 Infrared viewer. - The lens tissue paper, conforming to Specification MIL-P-13988, shall be taped securely over lenses with pressure sensitive tape (see 3.2.1). The infrared viewer headrest assembly shall be coated with talcum conforming to Specification MIL-P-1755. The receptacle protective caps shall be installed on all receptacles of the viewer. The viewer shall be mounted in place in the stowage box with neutral paper, conforming to Specification MIL-P-130, between viewer and pads of the stowage box (see figure 1).

3.4.1.3.4 Voltage regulator. - The receptacle protective caps shall be installed on both receptacles. The voltage regulator shall be placed in the stowage box and secured in position with the attaching hardware which will be used for installation on the vehicle (see figure 1).

3.4.1.3.5 High voltage cable assembly. - The connectors of the cable assembly shall be sealed with pressure sensitive tape (see 3.2.1). The cable shall be coiled and placed in the stowage box. The cable shall be immobilized by taping to infrared lamps. (See figure 1).

3.4.1.3.6 Spare infrared lamp lenses. - The lens tissue paper, conforming to Specification MIL-P-13988, shall be taped securely around the four spare lenses with pressure sensitive tape (see 3.2.1). The lenses shall be placed in the lens compartment as shown on figure 1.

3.4.1.3.7 Stowage box. - With all components fastened in place in the stowage box, the cover shall be closed and the hasps secured.

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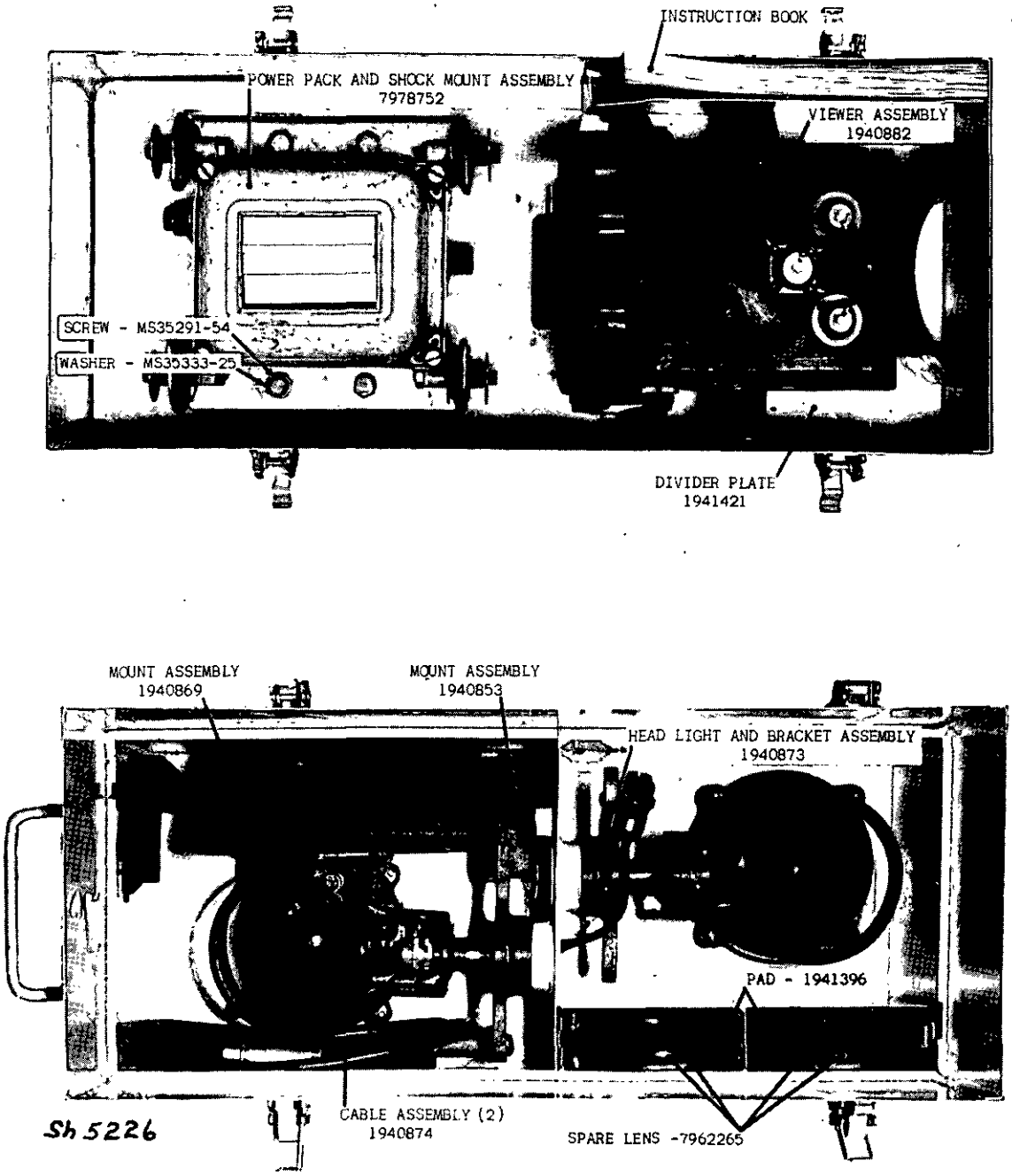


Figure 1 - Infrared Night Driving Kit Stowage Box.

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3.4.1.3.7.1 Cushioning. - Fiberboard pads fabricated from fiberboard, conforming to Specification PPP-B-636, shall be placed along sides and ends of stowage box for cushioning of clasps and handles.

3.4.1.3.7.2 Interior carton. - The padded stowage box shall be encased in a fiberboard box conforming to Specification PPP-B-636, type I or type II, class 1, using minimum 275 pound test board.

3.4.1.3.7.3 Bag. - Stowage box, cushioned and cartonized, shall be enclosed and sealed in a moisture vapor-proof bag conforming to Specification MIL-B-117, type I, class E.

3.4.1.3.7.4 Outer carton. - Bagged unit shall be encased in a fiberboard box conforming to type I or type II, class 2, compliance symbol V3S or V3C of Specification PPP-B-636. The fiberboard box shall be closed with pressure sensitive tape (see 3.2.1) in accordance with the appendix to Specification PPP-B-636.

3.4.2 Level B. -

3.4.2.1 Cleaning and Drying. - Cleaning and drying shall be as specified in 3.4.1.1 and 3.4.1.2.

3.4.2.2 Unit preservation and packaging. - The components of the infrared night driving kit shall be preserved and packaged as specified in 3.4.1.3.1 to 3.4.1.3.6 inclusive.

3.4.2.3 Stowage box. - The stowage box shall be packaged as described in 3.4.1.3.7 except for the following:

(a) The moisture vapor-proof bag shall be omitted.

(b) The outer fiberboard box shall be omitted.

3.4.3 Level C. - Packaging shall be such as to afford adequate protection against corrosion, deterioration and physical damage during shipment and interim indoor storage prior to use at the first receiving installation.

3.5 Packing. - Packing shall be level A, B or C as specified (see 6.1).

3.5.1 Level A. - The bagged and cartonized infrared stowage box shall be immobilized such that there shall be no pressure against the handles or hasps in a shipping container conforming to one of the following specifications at the option of the contractor:

PPP-B-601, Plywood box, overseas type, the most appropriate style and type for overseas shipment.

PPP-B-621, Nailed wood box, class 2, the most appropriate style for overseas shipment.

Box closure and strapping shall be as specified in the applicable box specification or appendix thereto.

3.5.2 Level B. - One bagged and cartonized infrared stowage box shall be immobilized such that there shall be no pressure against the handles or hasps in a shipping container conforming to one of the following specifications at the option of the contractor.

PPP-B-601, domestic type, the most appropriate style under domestic shipment.

PPP-B-621, class 1, the most appropriate style under domestic shipment.

Box closure shall be as specified in the applicable box specification or appendix thereto.

3.5.3 Level C. - The packaged infrared night driving kit shall be packed in containers of the type, size and kind commonly used for this purpose and in a manner which will assure both acceptance by the common carrier and safe delivery at destination at the lowest rate. Containers shall conform to the Uniform Freight Classification Rules and regulations or regulations of other carriers as applicable to the mode of transportation.

3.6 Marking. -

3.6.1 Interior Package. - Marking shall be applied on the barrier and marked in accordance with Standard MIL-STD-129.

3.6.2 Shipping container. - The shipping container shall be marked in accordance with Standard MIL-STD-129. Fragile markings shall be applied on each shipping container as specified in Standard MIL-STD-129. A designator arrow and the word "UP" will be stencilled on the container as specified in Standard MIL-STD-129.

3.6.3 Stock number. - The stock number, to be added in the identification data, shall be obtained from the contracting officer.

3.7 Workmanship. - All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedure specified herein shall be in accordance with the highest grades of practice associated with this type work.

4. QUALITY ASSURANCE PROVISIONS

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4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pilot pack. - A sample pilot pack shall consist of one completed package, packed as for shipment, and shall pass the quick leak test (4.3.3.1(a) and the rough handling test (4.3.3.2). The rough handling test shall precede the quick leak test. A pilot pack will not be required when such a pack has previously been tested and accepted for the same method for similar items by the same contractor, and satisfactory evidence can be furnished that the package being offered is identical to the approved pack. Pilot pack tests shall be repeated when changes in packaging and packing materials, processes or designs have been made.

4.2.1 Evaluation. - All materials used for the packaging and packing shall be free from breaks, tears, punctures or other damage which will affect the usefulness of the packaged equipment. The equipment within the package shall show no signs of damage and shall be free from all operational malfunctions.

4.3 Sampling for lot acceptance. -

4.3.1 Inspection lot. - All preserved, packaged and packed infrared night driving kits presented for delivery at one time shall be considered a lot for purposes of inspection in accordance with Standard MIL-STD-105.

4.3.1.2 Sampling for examination. - A random sample of the packaged containers shall be selected for the examination and tests specified in 4.3.2 and 4.3.3.1.

4.3.2 Inspection. -

4.3.2.1 Examination. - Each of the sample packaged containers selected by the inspector shall be opened and examined to determine compliance with this specification. Any sample packaged container failing to conform with this specification shall cause rejection of the lot represented by the sample.

4.3.3 Test procedures. -

4.3.3.1 Packaging tests. - The following packaging tests shall be conducted by the inspector:

a. Quick leak test. - A sample package box, selected from a lot which has passed the examination of 4.3.2.1 shall be subjected to the quick leak test as specified in Specification MIL-P-116 applicable to level A and B only.

b. Determination of cleanliness test. - For levels A, B and C packaging, the components of the infrared night driving kit shall be subjected to the determination of cleanliness test in accordance with Specification MIL-P-116.

4.3.3.2 Rough handling test. - For level A and B packing, the pilot pack only shall be subjected to the rough handling test in accordance with the requirements of Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable levels of preservation, packaging and packing (see 3.4 and 3.5).

6.2 Size and weight of packed shipping container. - The approximate inside dimension, cubage and gross weight for one packed shipping container for Level A are as follows:

Inside dimensions - 24 inches long x 10 inches wide x 12 inches high
Cubage - 1.66 cubic feet
Gross weight - 35 pounds

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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